

Eric Gitangu

Software Engineer — Payments Platforms, Distributed Systems & Security Architecture

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PROFESSIONAL SUMMARY

Software engineer with 11+ years designing, shipping and operating distributed, mission-critical systems — the last five in African fintech, mobility and energy access, where the systems move real money. Currently own platform architecture and data security for a mobility/fintech platform: five product surfaces and a mobile API behind one backend-for-frontend, 6,800+ unit tests, 280+ architecture decision records, an atomic release rail with automatic rollback, and card and mobile-money rails (Mastercard MPGS, M-Pesa, MTN, Tigo) built to PCI DSS SAQ-A scope with idempotent, fail-closed money paths. Earlier: CI/CD and quality gates for a PAYG solar platform across seven markets, an ML-security platform for medical-device software, and a digital-addressing pilot with a county government. Polyglot across TypeScript/Node.js, Java/Spring Boot, PHP, Python, Rust, Go and C#.NET; deep in event-driven architecture, resiliency and fault-tolerance patterns, public cloud and containers, observability and secure SDLC. Set architecture direction across teams, mentor engineers (15+), and stay in production code every day.

EXPERIENCE

Senior Full Stack Engineer — Platform Architecture & Data Security — Moovn Technologies

Feb 2026 – Present

Nairobi, Kenya (Seattle HQ)

Own architecture and data security for a mobility/fintech platform serving riders, drivers, corporates and partners across East and Southern Africa.

- Own the architecture of a money-moving platform: five Next.js surfaces and a mobile API behind one backend-for-frontend on a PHP monolith — 6,800+ unit tests and 280+ architecture decision records governing every change.
- Designed and operate the atomic release rail: immutable releases activated by a symlink flip, a post-flip smoke that decides healthy / roll back, a release guard on money-path files, and detached self-logging deploys — one command, under a minute, one-command rollback.
- Built card and mobile-money rails to PCI DSS SAQ-A scope: Mastercard Payment Gateway Services hosted sessions (card data never touches our servers), M-Pesa Daraja STK and B2C, MTN and Tigo — locality-locked rail resolution, fail-closed idempotency, signed and IP-allow-listed settlement callbacks.
- Designed a cross-product trust-score engine: HMAC-signed idempotent event ingestion from partner products, a tamper-evident hash-chained ledger, a budget-guarded nightly recompute over ~40,000 users, and a flag-gated phased rollout across admin, rider portal and mobile.
- Raised security posture end to end: key-only SSH with fail2ban, WebAuthn/TOTP MFA and reserved admin rights, audit hash-chains, gitleaks and blocking test gates in git hooks, purple-team pentest remediation, encrypted nightly off-box backups with restore drills.
- Instrumented a legacy application with Prometheus, Grafana and Loki without coupling app code; root-caused a failover storm to a mis-sequenced reboot. Author the team's release and deploy standards; mentor through structured review.

PHP TypeScript Next.js React Node.js MongoDB Redis Nginx Prometheus Grafana

Team Lead — QA & Engineering — Ignite Energy Access (formerly ENGIE Energy Access)

May 2025 – Jan 2026

Nairobi, Kenya

Owned CI/CD and quality infrastructure for a PAYG solar platform across seven African markets.

- Owned end-to-end CI/CD and QA infrastructure across seven markets (Kenya, Uganda, Tanzania, Rwanda, Mozambique, Côte d'Ivoire, Zambia) — testing standards, release gates and deployment workflows adopted across engineering.
- Built automated end-to-end suites (Playwright, Cypress) over payment processing, token generation, subscriptions and IoT provisioning; moved regression from multi-day manual cycles to automated runs on every pull request.
- Led the Rails 5 to 7.2 upgrade across the payment, PAYG token, IoT diagnostics, subscription and OTA firmware systems, spanning multi-tenant configurations for multiple markets.
- Designed asynchronous token generation (RabbitMQ) and a Redshift payment-event pipeline, removing synchronous bottlenecks under peak billing load; led and grew a team of QA engineers.

Ruby on Rails Playwright Cypress RabbitMQ Redshift PostgreSQL GitLab CI Docker Kubernetes K6

Director of Engineering / Lead Engineer — Vishnu Systems, Inc.

Jan 2024 – Aug 2025

Atlanta, GA · Remote

Founded and led engineering (concurrently with Baw Bab) for an ML security and robustness platform for software in and as medical devices (SIMD/SaMD).

- Built the platform from scratch: CNNs, RAG and GNNs to detect adversarial attacks on ML models, evaluate robustness and surface gaps against healthcare ML regulation (HIPAA, FDA 21 CFR Part 11, IEC 62304).
- Architected an AWS event-driven backend in Python and Rust (PyO3) with a Go middleware layer for routing, rate limiting and fan-out; circuit-breaker and back-off patterns kept a live degradation invisible to customers.
- Orchestrated ML pipelines with Airflow and tracked them end to end with MLflow — experiment versioning, model registry, reproducibility.

Python FastAPI Rust PyO3 Go AWS Kafka Airflow MLflow PyTorch

Lead Software Engineer — Baw Bab Technologies, Inc.

Apr 2023 – Aug 2025

Seattle, WA / Nairobi, Kenya

Led engineering (concurrently with Vishnu Systems) for a digital-addressing platform for last-mile delivery in informal urban areas across Africa.

- Built computer-vision pipelines over Azure, Google and OpenStreetMap satellite imagery to identify residential structures and convert them to Google Plus Codes at scale.
- Piloted the platform with the Nairobi County Government — the second digital-addressing programme of its kind in Africa — integrating with county land-registry records.
- Owned the full stack (React, Node.js, Kafka, MongoDB), the OpenAPI contracts and a Go backend-for-frontend separating field-agent mobile from county web; ran it on Azure, then hybrid Azure and GCP Cloud Run.

React Node.js Go Kafka MongoDB PostGIS Azure GCP Cloud Run OpenAPI

Senior Software Engineer — Reinsurance Group of America (RGA)

Jan 2022 – Jul 2022

Remote

- Refined service boundaries and data contracts with platform architects in a correctness-critical domain; defensive coding, integration testing, careful PostgreSQL migrations, OpenAPI and contract-test conventions for independent deploys.

Java Spring Boot PostgreSQL OpenAPI Pact

Independent Engineer & Mentor — Deveric Technologies

Jan 2019 – Feb 2022

Nairobi, Kenya / Remote

- Founded the Deveric mentoring initiative — structured mentorship for 15+ developers, 6+ advancing to mid or senior roles; earned 80+ technical certifications across cloud, backend, blockchain, AI/ML and security; shipped Go, Rust and Python projects in hackathons and open source.

Go Rust Python TypeScript AWS GCP Azure

Software Engineer — Vecima Networks Inc. (now Concurrent)

Jan 2018 – Jan 2019

Peachtree City, GA

- Wrote low-level C and embedded assembly for streaming pipelines with deterministic store-and-replay and zero frame loss; designed fault-tolerant delivery with redundant paths, graceful degradation and self-healing recovery.

C Assembly Linux CDN

Software Engineer — Veracode

Dec 2015 – Jan 2018

Burlington, MA

Application-security platform — integrating SAST/DAST/SCA into enterprise CI/CD and DevSecOps toolchains.

- Built integrations embedding security scanning into customers' CI/CD (Jenkins, GitLab CI, Azure DevOps); contributed to the mono-lith-to-microservices decomposition on Spring Boot, JPA and Hibernate with HATEOAS REST APIs.
- Owned the Visual Studio plugin (.NET / Windows) end to end, including the TLS 1.2 enforcement upgrade for regulated customers; every feature reviewed against the OWASP Top 10, aligned to NIST SSDF.

Java Spring Boot JPA/Hibernate C#/ .NET REST Jenkins OWASP

SELECTED PROJECTS

PesaCore — .NET 10 Cloud-Native Banking Platform · [pesacore-web-261341881894.africa-south1.run.app](#)

Core-banking platform on ASP.NET Core 10 with a Blazor BFF: CQRS with idempotent transfers, Postgres + Redis, OpenTelemetry to Prometheus/Grafana/Jaeger, hardened nginx edge — deployed serverless on GCP Cloud Run, 67 tests gating CI.

C# .NET 10 ASP.NET Core Clean Architecture MediatR CQRS Polly FluentValidation Entity Framework Core AutoMapper

LendStream · nanolend-app.vercel.app

Event-driven micro-lending reference platform — Java 21/Spring Boot and Clojure services over Kafka with CQRS, event sourcing and a transactional outbox; React/TypeScript front end.

Java 21 Spring Boot Clojure React TypeScript Kafka QStash PostgreSQL Redis Docker

UniCorns - Multi-Tenant SaaS Super-App · unicorns.run

Multi-tenant SaaS for African clinics, pharmacies and retail: eight Rust services on AWS Lambda, DynamoDB tenant isolation via Cognito claims, offline-first web/Android/iOS clients, four payment rails (M-Pesa, Airtel, Stripe, SHA FHIR claims) — ~2,450 automated tests.

Rust AWS Lambda DynamoDB Next.js Kotlin Claude AI M-Pesa

Vunachain — AgriTech Contract Farming & Traceability

Agricultural supply-chain compliance and traceability (EU deforestation regulation, Kenya Tea Act): Django/PostGIS geospatial checks, event-driven harvest pipeline anchoring provenance on Celo, Merkle-batched contract allocation, M-Pesa payouts. Demo-stage.

Rust AWS Lambda ARM64/Graviton EventBridge SQS DLQs S3 RDS PostgreSQL CloudWatch X-Ray

SwarmBench Multi-Agent LLM Empirical Study · developer.ericgitangu.com/pdf/research-paper.pdf

Empirical single-agent vs multi-agent LLM study authored as a Harbor benchmark: six iterations, two verifier types, codebases to 303K LOC — the negative result (no multi-agent wedge for tool-using agents) reported transparently.

Kimi K2.5 Fireworks AI Harbor Docker Python pandoc WeasyPrint LLM-as-judge Multi-Agent Systems Benchmark Design

AgentPlane — AI-Powered JIRA Board · agent-plane.ericgitangu.com

AI-augmented project management: a three-agent LangGraph pipeline decomposes an epic into estimated sub-tasks with a replayable decomposition audit log; FastAPI + Next.js, cost-bounded LLM calls.

Python 3.12 FastAPI LangGraph LangChain Anthropic Next.js 14 TypeScript TanStack Query CockroachDB MongoDB Atlas

CORE TECHNICAL SKILLS

Architecture & Systems

Distributed Systems · Microservices · Event-Driven · CQRS · Event Sourcing · Resiliency & Fault Tolerance · Hexagonal / Ports & Adapters · DDD · Multi-tenant · API Design

Payments, Security & Compliance

PCI DSS (SAQ-A) · Tokenization / hosted payment sessions · Idempotency & fail-closed money paths · Settlement & reconciliation · OAuth2/JWT · OWASP Top 10 · Secure SDLC / threat modelling · MFA (WebAuthn / TOTP) · Audit hash-chains · NIST SSDF

Backend & APIs

TypeScript · Node.js · Java · Spring Boot · PHP · Python · Rust · Go · C#/.NET · Kafka

AI/ML

TensorFlow · PyTorch · LLMs · OpenAI · Anthropic Claude · Google Gemini · RAG Systems · CNNs · GNNs · Transformers

Databases & Data

PostgreSQL · MongoDB · Redis · DynamoDB · MySQL · Kafka · RabbitMQ · Redshift · ETL/ELT Pipelines

Cloud & DevOps

AWS · GCP · Azure · Kubernetes · Docker · Terraform · CI/CD · Atomic releases / rollback · Prometheus / Grafana / Loki · OpenTelemetry

Frontend & Mobile

React · Next.js · TypeScript · Angular · Vue.js · Tailwind CSS · PWA · Kotlin + Jetpack Compose · Swift/SwiftUI · Electron

EDUCATION

Bachelor of Science in Computer Science · Minor: Mathematics — University of Massachusetts at Lowell

Sep 2011 – May 2015

• Microsoft Scholar and Google Scholar · Dean's List (nine semesters) · Omicron Delta Kappa

CERTIFICATIONS & OPEN SOURCE

- **Open Source & Certifications:** Public GitHub repositories across Rust, Go, TypeScript, Python, Java and C#/.NET (payments and lending platforms, distributed systems, ML pipelines); 80+ validated certifications across AWS, Azure, GCP, AI/ML, blockchain and application security.
- **Mentorship & Teaching:** Founded the Deveric mentorship initiative (2019–present): 15+ engineers mentored, 6+ advanced to senior or lead roles. CodePath.org volunteer instructor (2024), advanced algorithms cohort.